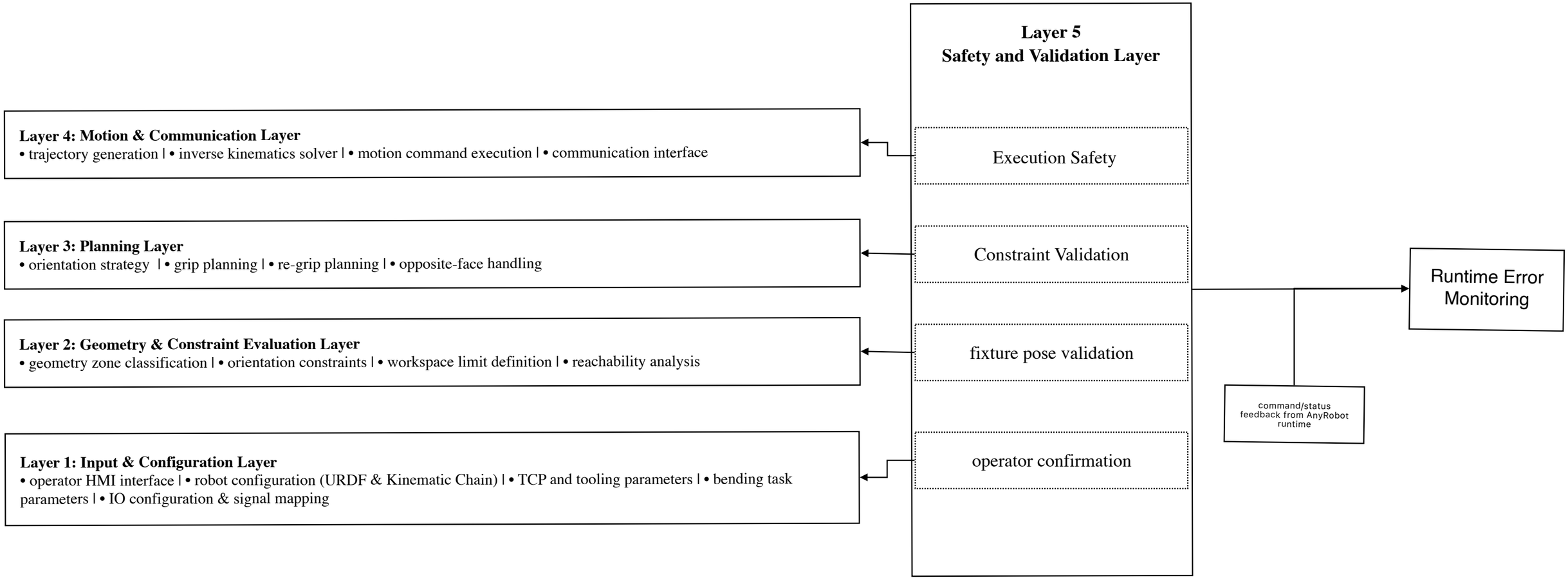
Layer 4: Motion & Communication Layer
• trajectory generation | • inverse kinematics solver | • motion command execution | • communication interface
Layer 3: Planning Layer
• orientation strategy | • grip planning | • re-grip planning | • opposite-face handling
Layer 2: Geometry & Constraint Evaluation Layer
• geometry zone classification | • orientation constraints | • workspace limit definition | • reachability analysis
Layer 1: Input & Configuration Layer
• operator HMI interface | • robot configuration (URDF & Kinematic Chain) | • TCP and tooling parameters | • bending task parameters | • IO configuration & signal mapping
Layer 5
Safety and Validation Layer
Execution Safety
Constraint Validation
fixture pose validation
operator confirmation
Runtime Error Monitoring
command/status feedback from AnyRobot runtime